Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*

Code: 105502 | *Subject: Database Management Systems* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which statement is FALSE?

(i) Every weak entity depends on a strong entity

(ii) A weak entity can have a partial key

(iii) Every many-to-many relationship becomes a separate relation

(iv) Every relation generated from ER always has a single candidate key

(b) For relation: R(A,B,C); FDs={A→B, B→A}, Which statement is true?

(i) A and B both are candidate keys

(ii) Only A is key

(iii) Only B is key

(iv) Neither A nor B is key

(c) The relational algebra expression $\pi_A(R) - \pi_A((\pi_A(R) \times S) - R)$ is equivalent to:

(i) Selection

(ii) Join

(iii) Division

(iv) Projection

(d) Which query is best solved using division operation?

(i) Find students enrolled in DBMS

(ii) Find students enrolled in all courses

(iii) Find students with marks >80

(iv) Find students from CSE

(e) A relation is in BCNF if:

(i) Every non-prime attribute depends on key

(ii) Every FD is trivial

(iii) Every determinant is a superkey

(iv) Every attribute is atomic

(f) Suppose relation R(A,B,C) is decomposed into: R1(A,B) and R2(B,C)
The decomposition is lossless if:

(i) B→A or B→C

(ii) A→B

(iii) A→C

(iv) C→B

(g) Compared with B-tree indexing, hashing is preferred when queries mainly involve:

(i) Range searches

(ii) Ordered traversal

(iii) Exact match search

(iv) Sorting

(h) For a B-tree of order m, every non-root internal node must have at least:

(i) m/2 children

(ii) [m/2] children

(iii) [m/2]−1 children

(iv) m−1 children

(i) Consider the schedule: R1(X), R2(X), W1(X), W2(X). The schedule is:

(i) View serializable only

(ii) Conflict serializable

(iii) Not serializable

(iv) Recoverable

(j) Under strict two-phase locking:

(i) Exclusive locks are held until commit/abort

(ii) Deadlocks are impossible

(iii) Locks can be released anytime

(iv) Transactions become serial

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Q.2 (a) Explain the three-level database architecture with a neat diagram. How does it support data abstraction and data independence? [7]

(b) Explain integrity constraints in DBMS. Discuss different types with examples. [7]

Q.3 (a) Design an ER diagram for a Hospital Management System containing Patients, Doctors, Departments, and Appointments. Clearly identify entities, attributes, relationships, and cardinalities. [7]

(b) What is a JOIN in SQL? Explain various JOIN operations with a suitable example for each and discuss their applications in database systems. [7]

Q.4 (a) Consider: [10]

Student(SID, Name, Dept, Age)
Course(CID, CName, Faculty)
Enroll(SID, CID, Marks)

1. Write relational algebra expressions to find:
   a) Students enrolled in DBMS
   b) Students scoring >80 marks
   c) Students not enrolled in any course
2. Write SQL queries:
   a) Find highest marks
   b) Find average marks department-wise
   c) Display students enrolled in more than two courses
3. Translate the following query into tuple relational calculus:
   "Find names of students from CSE department."
4. Convert the following relational algebra into SQL:
   $\pi_{Name}(\sigma_{Marks>85}(Student \bowtie Enroll))$

(b) For relation R(A,B,C,D,E) functional dependencies are given as follows: [4]

$A \rightarrow BC$
$CD \rightarrow E$
$B \rightarrow D$

Find candidate keys.

Q.5 (a) Explain different normal forms up to BCNF with examples. [7]

(b) Explain different indexing techniques used in databases. [7]

Q.6 (a) Consider a hash table of size 10 with the hash function h(k)=*k* mod 10. Insert the keys 23, 43, 13, 27, 37, and 50 into the hash table. Construct the hash table using separate chaining for collision resolution and then reconstruct it using linear probing. Compare the final arrangements obtained using both methods. [7]

(b) Define serializability. Explain conflict and view serializability. [7]

Q.7 (a) Explain two-phase locking protocol with an example. [7]

(b) Explain concurrency control problems such as lost update, dirty read, and unrepeatable read. [7]

Q.8 (a) Explain authentication, authorization, and the role of encryption in DBMS security. [7]

(b) Explain object-oriented databases and discuss their advantages over relational databases. [7]

Q.9 **Write short note on *any two* of the following:** [7 x 2 = 14]

(a) Dependency Preservation and Lossless Decomposition
(b) Database Recovery Techniques
(c) SQL injection attacks
(d) data mining

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